Assignment 1

SI 315

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Contents

1 Network 1 **1**

1.A Longest Cycles . . . 2

1.B Connectedness . . . 2

1.C Connected Components . . . 2

1.D Size of Connected Components . . . 2

1.E Breadth-First Search . . . 2

1.F Add One Edge . . . 3

2 Network 2 **3**

2.A Strong Triadic Closure Property . . . 3

2.B Fixing the Network . . . 4

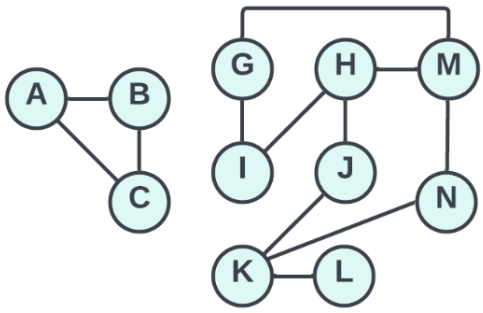
2.C Local Bridges . . . 4

2.D Average Local Clustering Coefficient . . . 4

3 Network 3 **5**

3.A Strong and Weak Ties . . . 5

1 Network 1



1.A) Longest Cycles

Problem

Identify the longest cycles in the network.

The cycle $I \to G \to M \to N \to K \to J \to H \to I$ has 7 nodes and is the only longest cycle.

1.B) Connectedness

Problem

Is the network connected? Explain.

The network is not connected because there are nodes that cannot be reached from other nodes (e.g. $B \to I$).

1.C) Connected Components

Problem

How many connected components does the network have? Which nodes are in each component?

The network has two connected components:

- Component 1: A, B, C
- Component 2: G, M, N, K, L, J, H, I

1.D) Size of Connected Components

Problem

How many nodes are in the largest component of the network?

The largest component is Component 2, which has 8 nodes: G, M, N, K, L, J, H, I.

1.E) Breadth-First Search

Problem

Find the distance from node H to every other node in the network using breadth-first search. Draw a diagram of the breadth-first search procedure similar to the one shown in class, where each new layer of nodes is identified and circled. Remember, the BFS diagram should have all of the nodes in it!

The breadth-first search layers starting at H are:

- Distance 0: H
- Distance 1: I, J, M
- Distance 2: G, K, N
- Distance 3: L
- Unreachable from H: A, B, C

1.F) Add One Edge

Problem

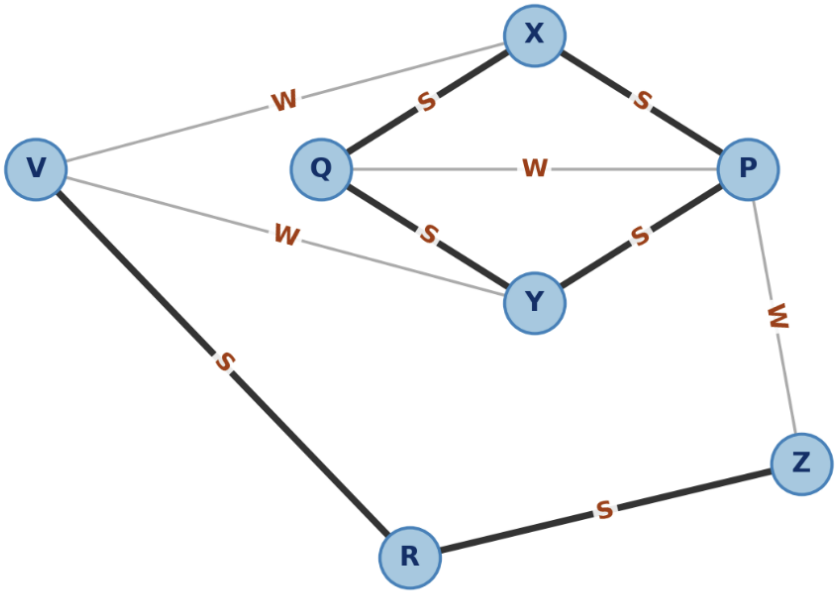
What is the longest cycle you can make by adding one edge to the network? Please specify what edge you would add, how long the longest resulting cycle is, and the cycle itself (i.e., the sequence of nodes). You do not have to explain how you found the edge and the cycle.

The longest cycle can be made by adding the edge $C \to G$. The resulting longest cycle is $I \to G \to C \to A \to B \to C \to G \to M \to N \to K \to J \to H \to I$, which has 11 nodes.

2 Network 2

Problem

In the network below, the ties (P, X), (P, Y), (Q, X), (Q, Y), (R, V), and (R, Z) are labeled strong (S), while all others are labeled weak (W).



2.A) Strong Triadic Closure Property

Problem

Identify all nodes that violate the Strong Triadic Closure Property. For each one, name its strong ties and the missing edge.

The nodes that violate the Strong Triadic Closure Property are P, Q, and R.

- P has strong ties to X and Y, but X and Y are not adjacent. The missing edge is XY.
- Q has strong ties to X and Y, but X and Y are not adjacent. The missing edge is XY.

- R has strong ties to V and Z, but V and Z are not adjacent. The missing edge is VZ.

No other node has two strong ties whose endpoints are not connected, so these are the only violations.

2.B) Fixing the Network

Problem

A student says that fixing this network requires the same number of new edges as the number of violations. Is the student right? Explain. Find a smallest set of edges such that adding them makes the network satisfy the Strong Triadic Closure Property. Explain why no smaller set of edges can work.

The student is wrong because there are 3 violations, but one edge can fix both violations for P and Q. The smallest set of edges that fixes all violations is $\{XY, VZ\}$. Adding XY fixes the violations at P and Q, and adding VZ fixes the violation at R. No smaller set of edges can work because each missing edge corresponds to a distinct violation that cannot be resolved by any other single edge.

2.C) Local Bridges

Problem

List all local bridges in the original network together with their spans. Then list all local bridges again after adding the edges from part (b). Explain why the list changed.

In the original network, the local bridges are:

- P–Z, span 4
- Z–R, span 4
- V–R, span 4

After adding the edges XY and VZ, the only local bridge is:

- P–Z, span 3

The list changes because the new edges create triangles and any edge that lies in a triangle cannot be a local bridge, since its endpoints already share a common neighbor.

2.D) Average Local Clustering Coefficient

Problem

Compute the average local clustering coefficient of the network before and after adding the edges identified in part (b). Does the direction of the change match what triadic closure predicts? Explain

Before adding the edges, the local clustering coefficients are:

$$C_P = \frac{2}{6} = \frac{1}{3}, \quad C_Q = \frac{2}{3}, \quad C_R = 0, \quad C_X = \frac{1}{3}, \quad C_Y = \frac{1}{3}, \quad C_Z = 0, \quad C_V = 0.$$

Thus,

$$\overline{C}_{\text{before}} = \frac{1}{7}\left(\frac{1}{3} + \frac{2}{3} + 0 + \frac{1}{3} + \frac{1}{3} + 0 + 0\right) = \frac{5}{21} \approx 0.238.$$

After adding XY and VZ, the local clustering coefficients become

$$C_P = \frac{3}{6} = \frac{1}{2}, \quad C_Q = 1, \quad C_R = 1, \quad C_X = \frac{4}{6} = \frac{2}{3}, \quad C_Y = \frac{4}{6} = \frac{2}{3}, \quad C_Z = \frac{1}{3}, \quad C_V = \frac{2}{6} = \frac{1}{3}.$$

So

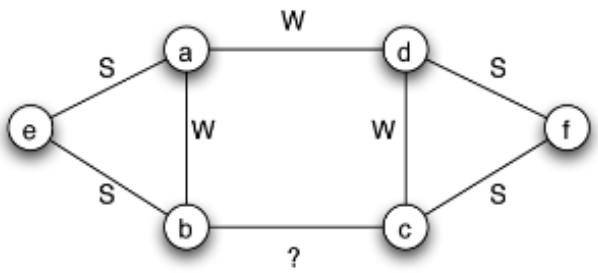
$$\overline{C}_{\text{after}} = \frac{1}{7}\left(\frac{1}{2} + 1 + 1 + \frac{2}{3} + \frac{2}{3} + \frac{1}{3} + \frac{1}{3}\right) = \frac{9}{14} \approx 0.643.$$

The average local clustering coefficient increases from about 0.238 to about 0.643. This matches triadic closure: adding edges that complete open triples creates more closed triplets and therefore raises clustering, so the direction of change is exactly what triadic closure predicts.

3 Network 3

Problem

This problem has 2 parts referencing the following network.



3.A) Strong and Weak Ties

Problem

Consider the graph in Figure 3.21, in which each edge — except the edge connecting b and c — is labeled as a strong tie (S) or a weak tie (W). According to the theory of strong and weak ties, with the strong triadic closure assumption, how would you expect the edge connecting b and c to be labeled? Give a brief (1-3 sentence) explanation for your answer.